FNCE30007 Tutorial 8 Dividends

Question 1

Consider a stock index currently standing at 250. The dividend yield on the index is 4% per annum, and the risk-free rate is 6% per annum. A three-month European call option on the index with a strike price of 245 is currently worth 10. What is the value of a three-month put option on the index with a strike price of 245, if each index point is valued at $25?

In this case $S_0 = 250$, $q = 0.04$, $r = 0.06$, $T = 0.25$, $K = 245$, and $c = 10$. Using put–call parity

$$c + Ke^{-rT} = p + S_0 e^{-qT}$$

or

$$p = c + Ke^{-rT} - S_0 e^{-qT}$$

Substituting:

$$p = 10 + 245e^{-0.25\times0.06} - 250e^{-0.25\times0.04} = 3.84$$

3.84*$25=$96

Question 2

An index currently stands at 696 and has a volatility of 30% per annum. The risk-free rate of interest is 7% per annum and the index provides a dividend yield of 4% per annum. Calculate the value of a three-month European put with an exercise price of 700, if each index point is valued at $25.

In this case $S_0 = 696$, $K = 700$, $r = 0.07$, $\sigma = 0.3$, $T = 0.25$ and $q = 0.04$.

$$d_1 = \frac{lnln\,(696/700)+(0.07-0.04+0.09/2)\times0.25}{0.3\sqrt{0.25}} = 0.0868 \approx 0.09 d_2 = d_1 - 0.3\sqrt{0.25} = -0.0632 \approx -0.06$$

and

$$N(-d_1) = 0.4641, \quad N(-d_2) = 0.5239$$

The value of the put, p, is given by:

$$p = 700e^{-0.07\times0.25} \times 0.5239 - 696e^{-0.04\times0.25} \times 0.4641 = 40.56$$

40.56*$25=$1,014

Question 3

Suppose that the spot price of the Canadian dollar is U.S. $0.95 and that the Canadian dollar/U.S. dollar exchange rate has a volatility of 8% per annum. The risk-free rates of interest in Canada and the United States are 4% and 5% per annum, respectively. Calculate the value of a European call option to buy one Canadian dollar for U.S. $0.95 in nine months. Use put-call parity to calculate the price of a European put option to sell one Canadian dollar for U.S. $0.95 in nine months. What is the price of a call option to buy U.S. $0.95 with one Canadian dollar in nine months?

In this case $S_0 = 0.95$, $K = 0.95$, $r = 0.05$, $r_f = 0.04$, $\sigma = 0.08$ and $T = 0.75$.

$$d_1 = \frac{ln\ ln\ (0.95/0.95) + (0.05 - 0.04 + 0.0064/2) \times 0.75}{0.08\sqrt{0.75}} = 0.1429 \approx 0.14$$

$$d_2 = d_1 - 0.08\sqrt{0.75} = 0.0736 \approx 0.07$$

and

$$N(d_1) = 0.5557, \quad N(d_2) = 0.5279$$

The value of the call, c, is given by

$$c = 0.95e^{-0.04*0.75} * 0.5557 - 0.95 * e^{-0.05*0.75} * 0.5279 = 0.0293$$

i.e., it is 2.93 cents. From put–call parity

$$p + S_0 e^{-r_f T} = c + Ke^{-rT}$$

so that

$$p = 0.0293 + 0.95 * e^{-0.05*0.75} - 0.95e^{-0.04*0.75} = 0.0224$$

The option to buy US$0.95 with C$1.00 is the same as an option to sell one Canadian dollar for US$0.95. This means that it is a put option on the Canadian dollar and its price is US$0.0224.

Question 4

Suppose that a portfolio is worth $60 million and the S&P 500 is at 1200. If the value of the portfolio mirrors the value of the index, what options should be purchased to provide protection against the value of the portfolio falling below $54 million in one year's time, assuming each option contract is for $100 times the index?

If the value of the portfolio mirrors the value of the index, the index can be expected to have dropped by 10% when the value of the portfolio drops by 10%. Hence when the value of the portfolio drops to $54 million the value of the index can be expected to be 1080. This indicates that put options with an exercise price of 1080 should be purchased. The options should be on:

$$\frac{60{,}000{,}000}{1200} = \$50{,}000$$

times the index. Each option contract is for $100 times the index. Hence 500 contracts should be purchased.

Question 5

Calculate u, d, and p when a binomial tree is constructed to value an option on a foreign currency. The tree step size is one month, the domestic interest rate is 5% per annum, the foreign interest rate is 8% per annum, and the volatility is 12% per annum.

In this case

$$e^{(0.05-0.08)\times 1/12} = 0.9975$$

$$u = e^{0.12\sqrt{1/12}} = 1.0352$$

$$d = 1/u = 0.9660$$

$$p = \frac{0.9975 - 0.9660}{1.0352 - 0.9660} = 0.4553$$

Question 6

Consider an option on a dividend-paying stock when the stock price is $30, the exercise price is $29, the risk-free interest rate is 5% per annum, the volatility is 25% per annum, and the time to maturity is four months. The stock is due to go ex-dividend in 1.5 months and the expected dividend is 50 cents.

a. **What is the price of the option if it is a European call? Check your answers using the simulator**
b. **What is the price of the option if it is a European put? Check your answers using the simulator**
c. **What happens to the price if the stock is expected to pay another 50 cents dividend in 6 months (ex-dividend date in 6 months)? Check your answers using the simulator. Comment on the findings.**

a. The present value of the dividend must be subtracted from the stock price. This gives a new stock price of:

$$30 - 0.5e^{-0.125 \times 0.05} = 29.5031$$

and

$$d_1 = \frac{ln(29.5031/29) + (0.05 + 0.25^2/2) \times 0.3333}{0.25\sqrt{0.3333}} = 0.3068 \approx 0.31$$

$$d_2 = 0.3068 - 0.25\sqrt{0.3333} = 0.1625 \approx 0.16$$

$$N(d_1) = 0.6217;\ N(d_2) = 0.5636$$

The price of the option is therefore

$$29.5031 * 0.6217 - 29 * e^{-0.05*0.3333} * 0.5636 = 2.2678$$

b. Because

$$N(-d_1) = 0.3783;\ N(-d_2) = 0.4364$$

the value of the option when it is a European put is

$$29 * e^{-0.05*0.3333} * 0.4364 - 29.5031 * 0.3783 = 1.2854$$

c. Your answer would stay the same as the next dividend will be paid after the option is expired.

Question 7

After reading the Virgin case, complete the following questions.

Note: The options used in this case are American-style options, whereas the standard Black–Scholes–Merton (BSM) formula applies to European options. For simplicity, we use the European BSM formula as an approximation to back out implied volatility from the observed

American option prices. In practice, an American-option pricing model would be more appropriate when the early-exercise feature is material.

Part A - Can One Anchor Volatility Price Other Calls?

(a) Calculate the BSM values using the same σ for all five calls.

(b) Compare the BSM values with the market mid-prices. Does one constant volatility reproduce all call prices exactly?

(c) What does this suggest about the constant-volatility assumption?

Answer:

(a) Use the 101C anchor volatility, σ = 52.22%, for every strike.

The resulting BSM values are:

Strike K	Market Mid	Common σ	BSM Value	Difference	Abs. % Error
95	10.080	52.22%	10.268	+0.188	1.86%
100	7.495	52.22%	7.542	+0.047	0.62%
101	7.065	52.22%	7.065	+0.000	0.00%
105	5.600	52.22%	5.379	-0.221	3.95%
110	4.215	52.22%	3.730	-0.485	11.51%

(b) No. The 101C is reproduced exactly because its own market price was used to infer the anchor volatility. The other strikes are not reproduced exactly. The largest error in this selected set occurs at K = 110, where the model value is 3.730 versus the market mid-price of 4.215.

(c) The result suggests that a single constant volatility does not describe the whole option-price cross-section very well. This motivates inferring a separate implied volatility for each strike and examining the volatility smile.

Part B - Call Smile and Put Smile

Now infer an individual implied volatility from each observed option mid-price. Use the call prices to construct a call smile and the put prices to construct a put smile. Both curves use the same strikes.

Answer:

7.1 Call Implied-Volatility Smile

Strike K	Call Mid-Price	Call Implied Volatility
80	21.800	54.93%
85	17.405	52.38%
90	13.435	50.93%
95	10.080	50.68%

100	7.495	51.87%
101	7.065	52.22%
105	5.600	53.88%
110	4.215	56.09%
115	3.185	58.17%
120	2.420	60.12%
125	1.840	61.84%
130	1.425	63.70%
135	1.110	65.41%
140	0.880	67.19%

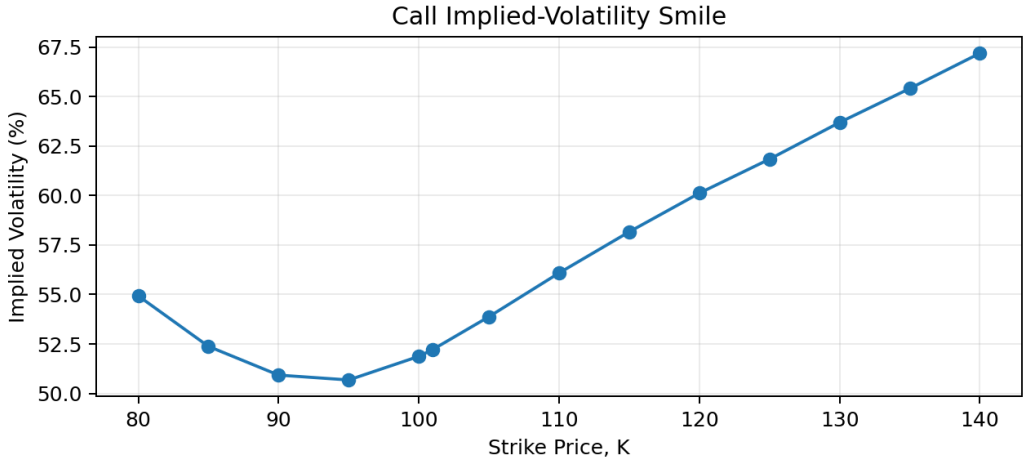


Figure 1. Call implied-volatility smile.

7.2 Put Implied-Volatility Smile

Strike K	Put Mid-Price	Put Implied Volatility
80	0.680	52.70%
85	1.285	51.14%
90	2.310	50.18%
95	3.970	50.37%
100	6.375	51.68%
101	6.945	52.07%
105	9.475	53.82%
110	13.100	56.28%
115	17.075	58.62%
120	21.300	60.76%
125	25.730	62.93%
130	30.310	65.18%
135	35.000	67.54%
140	39.760	69.87%

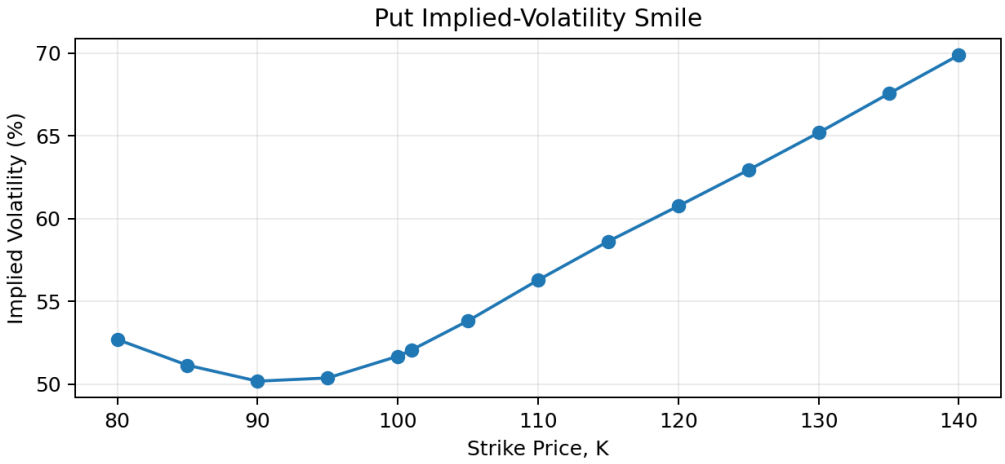


Figure 2. Put an implied-volatility smile.

7.3 What do both curves suggest about the BSM assumption that volatility is constant across strikes?

Answer: Both curves show that implied volatility is not constant across strikes. Call IV declines from about 54.93% at K = 80 to a minimum of about 50.68% around K = 95, then rises to about 67.19% at K = 140. Put IV declines from about 52.70% at K = 80 to a minimum of about 50.18% around K = 90, then rises to about 69.87% at K = 140. Therefore, both call and put prices imply a broadly similar smile-shaped relationship between strike and volatility, which is inconsistent with the strict BSM assumption of one constant σ across strikes.